

Response Letter with Annotations

Zhang, J., Brackbill, D., Yang, S., Becker, J., Herbert, N., & Centola, D. (2016). Support or competition? How online social networks increase physical activity: A randomized controlled trial. *Preventive Medicine Reports*, 4, 453–458. <https://doi.org/10.1016/j.pmedr.2016.08.008>

Annotations

1. Purposes

- a. Motivation (one-sentence summary)
 - To determine whether online social support or social comparison in social networks more effectively increases physical activity among young adults.
- b. Specific Rationale (several bullet points)
 - Physical inactivity is widespread among young adults and raises chronic disease and mortality risk.
 1. Nationwide, 69% of young adults are not meeting federal guidelines for physical activity
 - Online social networks are scalable, low-cost tools for health interventions, but it is unclear why and how they work.
 1. Prominent theoretical arguments state that social support and social comparison could be the main drivers for improving physical activity levels, but have not been explicitly tested
- c. Identify IV, DV, mediator, moderator
 - Independent variables
 1. Presence of social comparison
 - a. Competitive rankings
 2. Presence of social support
 - a. Teams, chat, team rewards
 - Dependent variable
 1. Total number of exercise classes attended over 11 weeks
 - a. 0 to 39 classes
 - Covariates (controls)
 1. Age, gender, race, department, prior gym membership, baseline physical activity
 - Moderator
 1. Team vs individual reward structure
- d. Research Questions and/or Hypotheses
 - No research questions are explicitly listed, but based off the reading here is what is implied
 1. Does social comparison within online social networks increase exercise/class attendance compared with no comparison?
 2. Does social support within online social networks increase exercise/class attendance compared with no support?
 3. Does combining social support and comparison produce additive or interaction effects on exercise behaviors and class attendance?

2. Method

- a. Research Design
 - A double-blind, four-arm randomized controlled trial (RCT) comparing social support vs social comparison vs combined vs control in an 11-week online exercise program
- b. Sample (representative sample?)
 - Convenience sample of graduate and professional students at Northeastern university (N= 790 participants).
 1. Not representative of the entire young adult population. Generalizability of this data is limited due to participants being from the same university.
 - Recruitment: University website, student email list, graduate association, flyers
- c. Measures
 - Primary outcome: Total number of exercise classes attended over 11 weeks, from instructor-verified class attendance records
 - Baseline measures:
 1. Demographics: gender, age, race, department, prior gym membership
 2. Physical status: endurance, strength, BMI from fitness evaluation
 3. Baseline physical activity via 3 CDC items, used to classify meeting 2008 guidelines or not
- d. Analysis
 - Power calculation: Planned N=688 (172 per arm) to achieve 90% power to detect small-medium effect
 - Initial analysis: ANOVA to test main effects of social support and social comparison on attendance
 - Primary Analysis: Multilevel regression model accounting for clustering within 6-person online networks
 1. Factors: social support, social comparison and their interaction
 2. Covariates: age, gender, race, department, prior gym membership

3. Results and Conclusions

- a. Answers to RQs and Hypotheses
 1. Does social comparison within online social networks increase exercise/class attendance compared with no comparison?
 - a. Yes, conditions with social comparison had 90% higher attendance than conditions without comparison
 - b. In multilevel models, social comparison increased attendance by 62% after covariate adjustment
 2. Does social support within online social networks increase exercise/class attendance compared with no support?
 - a. No, social support did not significantly affect exercise levels and had a non-significant negative effect in multilevel models
 - b. Support conditions' weekly attendance was slightly worse than control
 3. Does combining social support and comparison produce additive or interaction effects on exercise behaviors and class attendance?

- a. The combined group had the highest mean attendance across groups (38.5), but attendance was at similar rates to the social comparison group (35.7).
 - b. However, the support and comparison interaction was non-significant, so added support did not explain higher attendance; the success of the combined arm is attributed to team-based comparison alone
- b. Include Key Figures/Tables

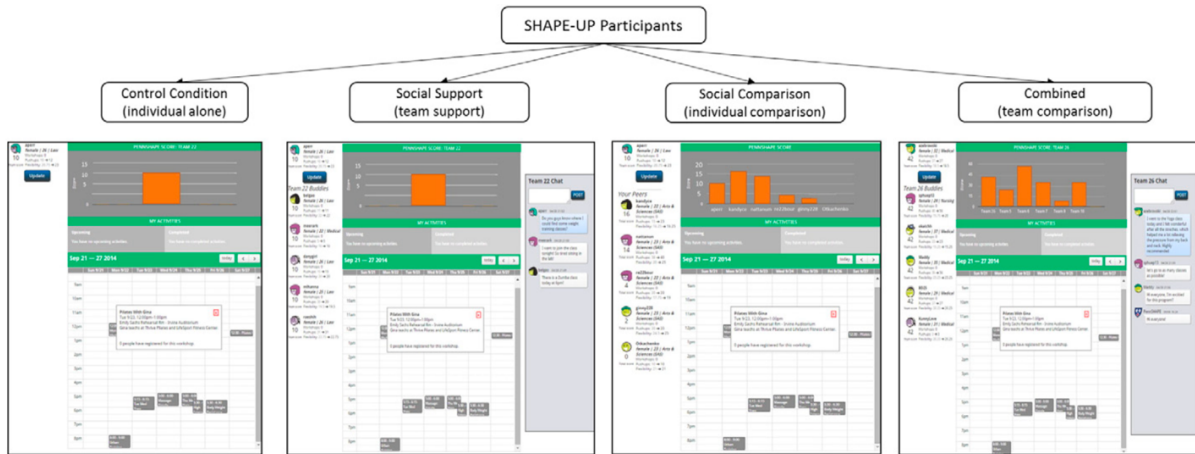


Fig. 1. Example webpage illustrations for the four experimental conditions in the trial, Philadelphia, PA, 2014.

Table 1
Intervention components in the four experimental conditions in the trial, Philadelphia, PA, 2014.

Intervention components	Control	Support	Comparison	Combined
SHAPE-UP website with an interactive calendar for class registration	x	x	x	x
Online networks with real-time web and email notifications of peer activities		x	x	x
Online chatting tool		x		x
Access to performance rankings of other peers or other teams			x	x
Rewards based on <i>individual</i> performance	x		x	
Rewards based on <i>team</i> performance		x		x

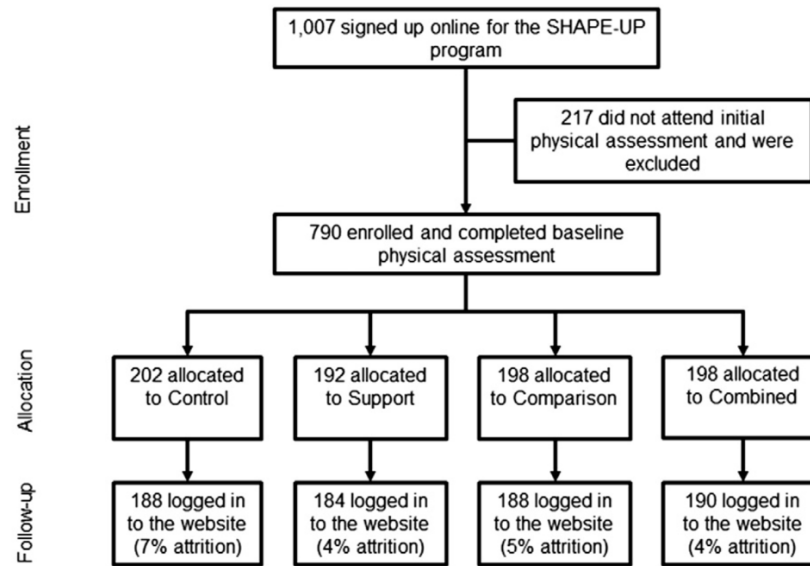


Fig. 2. Flow diagram of participants through the trial.

Table 2
Baseline demographic characteristics of participants per experimental condition, Philadelphia, PA, 2014.

Participants (N)	Total 790	Control 202	Support 192	Comparison 198	Combined 198	p values
Age (years; M [SD])	25.2 [3.4]	25.0 [2.7]	25.4 [3.5]	25.3 [3.8]	25.1 [3.5]	0.539 ^c
Male sex (N [%])	25.9	27.2	26.5	28.3	21.7	0.455 ^d
Body Mass Index (kg/m ² ; M [SD])	23.0 [3.8]	22.9 [4.0]	22.8 [3.6]	23.4 [3.8]	22.9 [3.7]	0.259 ^c
Overweight (BMI [25.0–29.9]; N [%])	124 [15.7]	26 [12.9]	30 [15.6]	37 [18.7]	31 [15.7]	0.465 ^d
Obese (BMI ≥ 30; N [%])	42 [5.3]	14 [6.9]	9 [4.7]	10 [5.1]	9 [4.5]	0.547 ^d
Met physical activity guideline (N [%]) ^a	336 [42.5]	87 [43.1]	80 [41.7]	83 [41.9]	85 [42.9]	0.998 ^d
Race (N [%]) ^b						0.448 ^d
White	352 [44.6]	88 [43.5]	85 [44.3]	100 [50.5]	79 [39.9]	
Black	58 [7.3]	14 [7.9]	12 [6.2]	13 [6.6]	19 [9.6]	
Hispanic	62 [7.8]	15 [7.4]	12 [6.3]	12 [6.1]	23 [11.6]	
Asian	287 [36.3]	73 [36.1]	76 [39.6]	61 [30.8]	77 [38.9]	

^a Participants met the guideline if they engaged in strength-building activity on at least 2 days and engaged in either 20 min of vigorous activity on at least 4 days or 30 min of moderate activity on at least 5 days.

^b The omitted race category is "Other."

^c The p values were based on one-way analyses of variance.

^d The p values were based on chi-squared tests.

Table 3

Multilevel models for the effects of experimental conditions on exercise class attendance, Philadelphia, PA, 2014.

	Class attendance	
	Unadjusted for covariates Estimate (95% CI)	Adjusted for covariates Estimate (95% CI)
Comparison	0.97* (− 0.06, 2.00)	1.06** (0.08, 2.04)
Support	− 0.17 (− 1.21, 0.88)	− 0.21 (− 1.21, 0.79)
Comparison × support	0.18 (− 1.38, 1.74)	0.06 (− 1.43, 1.55)
Constant	1.19*** (0.56, 1.82)	1.72 (− 1.38, 4.82)
Observations (N)	790	789
Log likelihood	− 2,309.55	− 2,271.03
Akaike Inf. Crit.	4,631.10	4,594.05
Bayesian Inf. Crit.	4,659.14	4,715.49

Notes: Covariates included age, gender, race, department, and having a gym membership in the previous semester. The difference in sample sizes arises from one missing data point for age.

* Boldface indicates statistical significance at $p < 0.1$.

** Boldface indicates statistical significance at $p < 0.05$.

*** Boldface indicates statistical significance at $p < 0.01$.

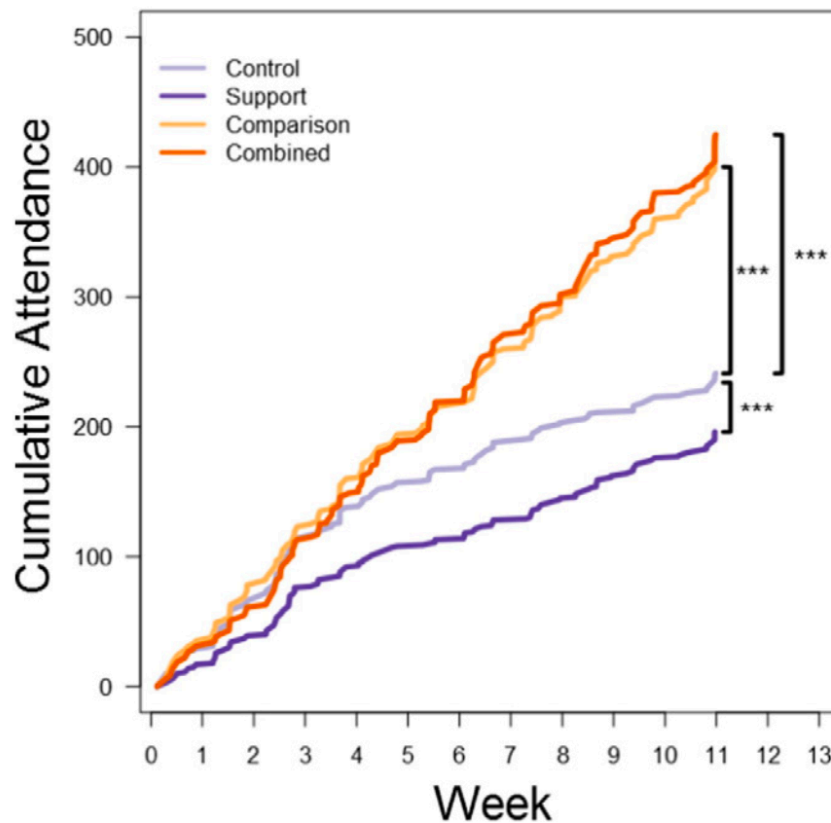


Fig. 3. Cumulative attendance at exercise classes in each of the four conditions.
 *** $p < 0.001$.

c. Authors' Conclusions

- Authors conclude that social comparison in online networks was a significantly stronger motivator of physical activity than social support. Furthermore, individual vs team incentives did not explain effects; instead, both individual and team rewards were equally effective when comparison was present. Adding a minimal competitive reference point to a supportive environment transformed the online network into a highly motivating setting.

4. Discussion

a. Implications

- Health programs, online fitness platforms, and peer communities can intentionally build competitive comparison to enhance exercise engagement and adopting and maintaining other health behaviors.

b. Limitations

- Networks were artificially constructed. Participants were randomly grouped, unlike naturally formed friendship networks, so effects of social support may be underestimated in this study
- No post-intervention psychosocial measures (perceived support, norms, comparison), so theoretical mechanisms and mediation could not be tested

- Participants within this study are all graduate and professional students at Northeastern University. This means that results may not be generalizable across all young adults

c. Future Directions

- Replicate this design in other populations and settings and with longer-term outcomes (e.g., weight, strength)
- Add mediation surveys and process evaluation to quantify psychological mechanisms and engagement patterns
- Test whether constructed comparison networks are similarly powerful for other health behaviors beyond physical activity like medication adherence, diabetes control, smoking cessation, vaccinations, weight loss, and medical screenings.

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Response Letter (~600 words)

- Your assessment of the study

This study was fascinating and compelling to me due to my interest in health communications and how they utilized networks from an experimental standpoint. The first aspect that was incredibly strong was the theoretical backing of this work. Within the introduction they clearly laid out how previous studies analyzing online social networks increasing effects on physical activity theoretically happen due to social comparison or social support. As the introduction moves forward, they provide a clear narrative to how both social comparison and social support have a substantial theoretical rational to why they would be effective within social networks focusing on physical activity. This section clearly provides a narrative and the “so what?” behind this work. Beyond this it gets you excited about the work being presented and what’s about to be found.

Another aspect that was notable within this work was the methodology and research design. This randomized controlled trail was strengthened by the use of computer-generated number sequencing to construct the social networks ensured randomization of experimental vs. control group placement. Additionally, the SHAPE-UP website was incredibly comprehensive and supported the progression of this study. This website managed various logistics that made this study realistic for participants and more feasible for researchers as it handled class registration, send web and email notifications when peers attended classes, tracked individual and team attendance, and displayed performance rankings and leaderboards. Furthermore, using instructors to track attendance was a fine-tuned way to track attendance for without the fear of self-reporting bias from participants.

- Weaknesses

One of the major weaknesses of this study, which was spoken about in the limitations, was the non-representative sample used. This population of graduate and professional students at Northeastern University is not a fair representation of the entire young adult population in the United States. To start this population may have more financial accessibility to gym setting or physical activities on campus due to student fees or student discounts at gyms off campus. Additionally, due to programs being held on campus, these participants have geographic advantages and access to gyms that most young adults may not have. In short this was a convenience sample and that raises concerns about the generalizability of results.

Another weakness I personally noticed was within the future directions of this study. Authors suggest that this study’s design should be replicated and evaluate other long-term health topics. While I understand the rationale behind this based on the current findings, this approach to health may not be effective and potentially have backfiring effects. For example, with weight loss, comparing one’s weight loss with other may increase feelings of accomplishment, however it can also ignite self-blame and defensiveness if weight loss goals aren’t accomplished or comparable to peers. This can then lead to resisting or reasoning away exercise altogether to avoid comparison and negative feelings about oneself. While social comparison may be effective in inherently competitive contexts such as exercise, health is very personal. Not everything should be socially compared as it can make individual health progress more difficult and stressful in the long run.